**VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES – TECHNICAL CAMPUS**

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**DATE:** October 24, 2026

**REF:** RESUB-2026-ARES-01

**To: Prof. Amir Hussain**, Editor-in-Chief, *Cognitive Computation*, Springer Nature
**Cc: Mr. Vincent Salvo**, Journals Editorial Office (JEO) Assistant, Group Publishing Operations, Springer Nature (vincent.salvo@springernature.com)

**SUBJECT: Submission of New Manuscript Following Editorial Guidance on Previous Submission ID: eda35bea-a77d-4e7d-ba30-edb38c688296**
**Manuscript Title: *ARES: Autonomous Real-Time Evaluation System — An AI-Driven Multi-Modal Truth Infrastructure for Misinformation Defense***

Dear Prof. Hussain and Mr. Salvo,

I am writing to formally submit our research manuscript, titled **"ARES: Autonomous Real-Time Evaluation System — An AI-Driven Multi-Modal Truth Infrastructure for Misinformation Defense"**, for consideration as a **New Submission** in *Cognitive Computation*.

## Editorial Guidance & Context of Resubmission

This submission directly follows formal correspondence with Mr. Vincent Salvo of the Springer Nature Journals Editorial Office regarding our previous submission, **Manuscript ID: eda35bea-a77d-4e7d-ba30-edb38c688296**. Following our follow-up inquiry dated November 16–17, 2025, regarding the rejection decision and options for manuscript reconsideration, Mr. Salvo explicitly advised on November 17, 2025:

> *"Dear Dr. Singh, Thank you for your response. As your previous manuscript has been rejected, you may submit your updated manuscript as new submission in the system. If you have further concerns, please let me know. Kind Regards, Vince (Vincent Salvo, Journals Editorial Office Assistant, Springer Nature)"*

We fully acknowledged that the previous rejection was primarily due to severe technical, formatting, and LaTeX rendering defects in the submitted files—specifically, overflowing mathematical expressions, unrendered citation placeholders ([?]), raw markup leaks, misaligned plots, and the lack of accessible code repositories at the time of submission. Importantly, the Reviewers also affirmed the core novelty, architectural vision, and practical relevance of the multi-modal truth infrastructure.
In accordance with Mr. Salvo's guidance, we have submitted this fully overhauled, publication-grade manuscript as a **New Submission**, accompanied by an exhaustive, 7-page **Point-by-Point Response to Editor and Reviewers** document (ARES_response_to_reviewers.pdf). Rather than offering superficial cosmetic edits, we undertook a ground-up technical, empirical, and mathematical overhaul of the entire research platform across 16 core modules.

## Key Pillars of the Comprehensive Overhaul & New Computations

The revised manuscript—spanning exactly 43 camera-ready pages in standard double-column publication format—incorporates the following rigorous scientific enhancements and new computational evaluations:

- **Complete Typesetting & Mathematical Rebuild:** Every mathematical formulation (adversarial loss, ingestion queueing models, and bounded manifold mutation derivations) has been re-typeset using standard two-column macros with explicit line breaking. All raw LaTeX listings (Page 34) and unrendered Mermaid markup (Page 35) have been replaced with 300 DPI vector graphics (**Figure 38** terminal trace and **Figure 37** system hierarchy). An automated scan confirms **zero margin overflow, formula clipping, or text collisions across all 43 pages**.
- **Ethical Course-Correction & Grounded Benchmarking:** In strict compliance with research ethics, we completely eliminated ungrounded narrative claims of operational societal intervention (e.g., traffic cuts during floods, military alerts, live post takedowns, and public trust surveys). All operational scenarios have been reframed into controlled, reproducible evaluations on our newly engineered **ARES-Synth v1** benchmark suite (1,858 verified items), explicitly designated **[SIMULATED BENCHMARK EVALUATION]**, and retrospective verification traces on frozen historical fact-checking corpora (CASE-01 to CASE-05). Section XIII-E explicitly discloses that human trust surveys require formal Institutional Review Board (IRB) approval.
- **Complete De-Branding & Standalone Scientific Integrity:** All internal development milestone names, phase labels, and auditing markers have been excised. Headers and footers have been cleaned to present a pristine, standalone academic monograph adhering to the highest standards of scientific computing.
- **Physical Hardware Telemetry & Latency Profiling:** All benchmarks were executed on a dedicated commodity workstation: **NVIDIA GeForce RTX 4060 Laptop GPU** (8GB GDDR6 VRAM, 3072 CUDA cores), **AMD Ryzen 7 7840HS CPU**, and **32GB DDR5 RAM**. Verified single-claim sequential latency is bounded between **14.8–27.2 ms/claim**, with asynchronous batched worker throughput reaching **3,560 requests/second** at a minimal **2,352 MB** VRAM footprint (**Table XX**, Page 17).
- **Statistical Posterior Calibration via Temperature Scaling:** Implemented post-hoc Platt / Temperature scaling (T = 1.34), reducing Expected Calibration Error from 0.1420 down to **ECE = 0.0312** across 13 public test sets, verified via dual-panel reliability curves and Brier score tracking (**Figure 16**, Page 9; **Table 07**, Page 36).

• **Vision & Speech Deepfake Empirical Portfolios:** Cross-dataset transfer from FaceForensics++ to Celeb-DF v2 (2,030 test videos, 98.4% fake prevalence) achieved video-level **ROC-AUC = 0.7184 [95% CI: 0.702, 0.734]** (**Figure 11-B**). Speech spoofing on ASVspoof 2019 LA with silence shortcut controls achieved **EER = 2.41%** and **Macro-F1 = 0.9760** (**Section XXXI-C**).

• **13-Dataset Multi-Baseline Progression & 10-System Matrix:** Multi-baseline comparisons across 13 public benchmarks show systematic progression from TF-IDF (0.5303) to Full ARES 2.0 (**Macro-F1 = 0.9010 [95% CI: 0.892, 0.910]**) supported by paired bootstrap tests (B = 10,000), exact McNemar tests, and Holm-Bonferroni corrections (**Table XXXVII**, Page 40). **Table XXI** (Page 17) benchmarks ARES against 10 state-of-the-art verification systems across 8 functional dimensions.

• **Full Public Open-Source Reproducibility:** Public GitHub repository under Apache-2.0 at https://github.com/gurarpitzz/Ares-Cognitive-Computation (Release Tag: v2.0.0). Complete codebase, models, and frozen dataset manifests archived with permanent DOI: **10.5281/zenodo.10826492**. Complete test suite of **110 passing unit tests** executable via pytest ares_v2/tests -q in under 45 seconds.

• **Comprehensive Citation Reconciliation:** Zero inline [?] placeholders remain. The bibliography was verified and expanded to 65 verified peer-reviewed publications with active DOIs.

## Accompanying Submission Files & Mandatory Declarations

**Accompanying Files:** (1) **Main Technical Manuscript:** ARES_Cognitive_Computation_Revised.pdf (Exactly 43 pages, standard double-column format). (2) **Point-by-Point Response to Reviewers:** ARES_response_to_reviewers.pdf (Detailed rebuttal mapping every comment to revised sections). (3) **Open-Source Code & Data Manifests:** Publicly accessible via the Zenodo and GitHub repositories referenced above. **Mandatory Declarations:** • **Originality:** The manuscript represents original work that has not been published previously and is not under consideration elsewhere. • **Conflict of Interest:** The author declares that there are no financial, commercial, institutional, or personal conflicts of interest. • **Research Ethics:** The study relies strictly on public benchmark datasets and synthetic perturbation simulations. No unauthorized operational deployments or human subjects experiments were conducted. • **Open Access Agreement:** The author confirms full agreement with Springer Nature's publishing policies and Open Access licensing terms (CC BY 4.0).

We trust that the exhaustive improvements, extensive empirical additions, and complete resolution of all previous concerns will allow the Reviewers and the Editorial Board to evaluate the scientific contributions of ARES on their true merits. Thank you very much for your time, consideration, and continued editorial guidance.

Respectfully submitted,

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